Iteration and simulation

Warm-up question

- A roulette wheel has 38 slots numbered 00, 0, and 1–36. Two are green, 18 are red, and 18 are black.
- If a gambler bets based on color, the return on a $1 bet is $2
- A gambler has $50, and will continuously bet $1 on red until they double their money (have $100) or lose the money they came with
- What is the probability the gambler doubles their money?

Question: Without calculating probabilities, how could you design an experiment to estimate this probability?

Designing an experiment

Step 1: representing the roulette wheel

```r
wheel <- c(rep("green", 2), rep("black", 18), rep("red", 18))

wheel
```

```
 [1] "green" "green" "black" "black" "black" "black" "black" "black" "black"
"black"
[10] "black" "black" "black" "black" "black" "black" "black" "black" "black"
"black"
[19] "black" "black" "red"   "red"   "red"   "red"   "red"   "red"   "red"
"red"
[28] "red"   "red"   "red"   "red"   "red"   "red"   "red"   "red"   "red"
"red"
[37] "red"   "red"
```

- `rep` repeats a value a specified number of times
- `c()` combines vectors into a single vector

Step 2: spin the wheel!

```
1 spin <- sample(wheel, size = 1)
2
3 spin
```

```
[1] "red"
```

Step 3: change in money

```
money <- 50
spin <- sample(wheel, size = 1)

if(spin == "red"){
  money <- money + 1
} else {
  money <- money - 1
}

spin
```

```
[1] "black"
```

```
money
```

```
[1] 49
```

- if the result was red, gain a dollar
- otherwise, lose a dollar

Step 3: change in money

Another way of writing the conditional statement:

```
money <- 50
spin <- sample(wheel, size = 1)

money <- ifelse(spin == "red", money + 1, money - 1)

spin
```

```
[1] "black"
```

```
money
```

```
[1] 49
```

Step 4: keep spinning

The gambler continues to bet until they have $0 or $100.

Question: Is a `for` loop appropriate for iterating the betting process?

Step 4: keep spinning

```r
money <- 50 # starting money

while(money > 0 & money < 100){
  spin <- sample(wheel, size = 1)
  money <- ifelse(spin == "red", money + 1, money - 1)
}

money
```

```
[1] 0
```

- `while` loop: repeat the process until the condition is true

Step 5: repeat the process

```r
set.seed(279)
nsim <- 1000
results <- rep(NA, nsim)

for(i in 1:nsim){
  money <- 50 # starting money

  while(money > 0 & money < 100){
    spin <- sample(wheel, size = 1)
    money <- ifelse(spin == "red", money + 1, money - 1)
  }

  results[i] <- ...
}
```

- What should I check at each iteration?

Step 5: repeat the process

```r
set.seed(279)
nsim <- 1000
results <- rep(NA, nsim)

for(i in 1:nsim){
  money <- 50 # starting money

  while(money > 0 & money < 100){
    spin <- sample(wheel, size = 1)
    money <- ifelse(spin == "red", money + 1, money - 1)
  }

  results[i] <- money == 100
}

mean(results)
```

```
[1] 0.008
```

Class activity

- Work independently or with a neighbor on the class activity
- At the end of class, submit your work as an HTML file on Canvas (one per group, list all your names)